

SAD

**MAKERERE UNIVERSITY BUSINESS SCHOOL,
FINAL EXAMINATIONS FOR BACHELOR OF BUSINESS COMPUTING
OF MAKERERE UNIVERSITY, ACADEMIC YEAR 2022/2023**

COURSE NAME: SYSTEMS ANALYSIS AND DESIGN

YEAR OF STUDY : ONE

COURSE CODE: BUC1229

SEMESTER : TWO

DATE: Sunday, June 04, 2023

TIME : 02-05:00 PM

INSTRUCTIONS

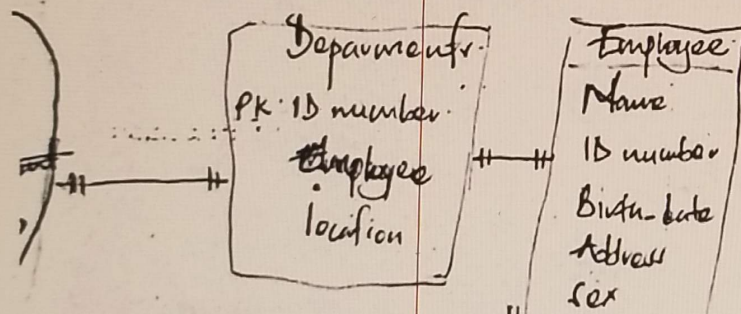
1. **ANSWER** any **FOUR** questions
 2. All questions carry equal marks (25 marks each)
 3. Clear handwriting, illustrations, and examples will be an added advantage
 4. Do not write anything on this question paper
 5. Each question **MUST** start from a new page
- Other instructions are provided on the answer booklet

Question One

UWED is an organization made up of various departments, each having a name, identification number, and an employee who is the manager. A department may be located in different locations. Information about an employee includes name, identification number, birth date, address, and salary. Each employee is assigned to one department. The date the manager is appointed to a department is also tracked. Employees may be directly supervised by another employee on a project within the organization is controlled by a department. Employees (not necessarily the controlling dept.) are assigned to projects. Information about projects includes project ID, and location. Hours spent by employees on each project are also kept.

You are required to:

- Identify the main entity types *Department, employee, Project* (03 marks)
- Identify attributes that make up the entity types (06 marks)
- Identify the relationship that exists between the entity types (03 marks)
- Determine the primary key attributes for each entity type. *One to one* (03 marks)
- Draw an ER diagram to represent the data requirements above *ID no Project ID* (10 marks)



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Question Two:

Information consists of data that has been organized to help answer questions and solve problems so, the purpose of an information system is to turn raw data into useful information that can be used for decision-making in an organization.

- a) Define the term information systems (6)
- b) Briefly describe a complete computer-based information system in terms of its components (13)
- c) Discuss the activities performed by an information system (10)

Question Three:

A stakeholder is any person who has an interest in a proposed information system. Stakeholders include both technical and non-technical workers. They can also be the internal or external stakeholders. Whatever their roles in an organization are it doesn't matter.

- a) Explain the roles of any five (5) stakeholders in the information system development process (10 marks)
- b) Explain the qualities of a modern systems analyst in information system development (12 marks)
- c) Explain any three factors that must be taken into consideration when developing an information system (03 Marks)

Question Four:

To design and implement an information system, there is a need to have a methodological approach in order to meet the time against physical and human constraints. A system development methodology refers to the steps that are used to form, plan, and control the process of developing an information system.

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a) Using a well-labeled diagram discuss the 4 steps involved in a prototyping process.

(08)

b) Discuss the advantages of the prototype approach to system development

(05)

c) What are the key factors to consider when choosing a system development methodology for a particular project?

(12 marks)

Question Five:

Software requirement analysis is the software engineering practice that, at the top level, software architecture, translates stakeholder needs and expectations into a viable set of requirements.

-) Explain the term "Problem" in three different perspectives in information system development (03 marks)
-) Discuss the different types of software requirements as used in information system development (10 marks)
-) PIECES framework is used to classify identified requirements into six subject areas: Performance, Information, Economy, Control, Efficiency, and Services. Discuss (12 marks)

Question Six:

The main aim of fact-finding techniques is to determine the information requirements of an organization by analysts to prepare a precise System Requirement Specification understood by the users.

-) Define the term Data (03 marks)
-) Discuss any Five(5) data collection techniques used in systems Analysis (12 marks)
-) Show the advantages and disadvantages of each technique in (b) above (10 marks)

...END....